CiteGuard: Evidence-First Detection, Verification and Explanation of Fabricated Citations with LLMs

Miten Gandhi
Introduction to Large Language Models (ILLM) — Final Project
https://github.com/mjgandhi2305/ILLM_FinalProject_Citeguard

November 2026

Abstract

Large language models (LLMs) now help write scientific text, and they invent references that look real: plausible titles, real-sounding authors, real venues, even real DOIs that belong to other papers. Detecting such citations sounds like a natural job for an LLM, but an LLM judging a citation from memory is itself prone to hallucination and, as we show, wrongly rejects correct citations it does not recognise. We present CiteGuard, a citation verifier built on an *evidence-first* principle: whether a cited work exists is decided only by retrieval from scholarly indexes (doi.org, Crossref, OpenAlex, Semantic Scholar, arXiv, PubMed) and deterministic record linkage, while the LLM is confined to tasks that evidence can check: parsing messy references, choosing among retrieved candidates, rewriting failed searches, writing explanations whose every claim must cite evidence, and an opt-in check of whether the cited abstract supports the citing sentence. CiteGuard assigns one of five verdicts (verified, verified with errors, misattributed, fabricated, unverifiable), explains it, and proposes a corrected citation or real alternatives. We also build CiteBench, a benchmark of real, perturbed and fabricated citations, and evaluate with a held-out protocol in which every split is scored once before its errors are examined. On a fresh held-out split of 150 citations, the deterministic pipeline reaches 0.960 five-class accuracy and 1.000 fake-detection F1 while calling 0.0% of correct citations fake. The strongest LLM-only baseline (`gpt-oss-120b`) reaches 0.727 F1 and calls 13.3% of correct citations fake, and a DOI-lookup baseline detects none of the fabrications. LLM explanations pass a grounding validator in 19 of 19 cases, and the claim-support check flags 92% of mismatched claims with no false alarms on 25 papers.

1 Introduction

Citations are how a paper hands its claims off to evidence. When a writer asks an LLM for "related work with references", the model produces citation-shaped strings, and a sizable share of them do not exist. Walters and Wilder [29] found that 55% of the references produced by GPT-3.5 and 18% of those produced by GPT-4 in short literature reviews were fabricated, and that many real references carried substantive errors; medical-domain studies report similar rates [2–4]. These references now reach the record: an audit of 111 million references in arXiv, bioRxiv, SSRN and PubMed Central estimates 146,932 non-existent citations in 2025 alone [30], and hallucinated references have been found in accepted NeurIPS 2025 papers [13]. Reviewers cannot check every reference by hand, so automatic verification has become part of the review process.

The obvious tool is an LLM: ask it whether a reference is real. This fails for a structural reason. A model that judges a citation from its parametric memory is subject to the same hallucination it is asked to detect [10, 11]. It recognises famous papers and distrusts everything else, so it flags most fabrications only because it also flags many obscure real papers. HALLMARK [13] measured exactly this on BibTeX entries: zero-shot LLM verifiers span an order of magnitude in false-positive rate (FPR), from 4% to 70%, and giving them database lookups as tools raised recall but also raised FPR. Because every flag costs a human review, the FPR on real citations, not recall, decides whether a verifier is usable. Retrieval-only checkers have the opposite weakness. A DOI resolver cannot see a fabricated title that carries a real DOI, and a binary found/not-found search conflates "not indexed" (books, theses, recent or non-English work) with "does not exist", and gives the user nothing to act on.

Our insight is that existence is a question of *evidence*, not of *judgement*, and that the LLM is most useful exactly where its output can be checked against that evidence. We build CiteGuard around this division of labour (fig. 1). Identifiers are extracted by regex and resolved; the reference is searched in several scholarly indexes; a calibrated record-linkage model scores each candidate; and deterministic rules map the field-level comparison to a verdict. The LLM parses references that rules cannot, adjudicates only among retrieved candidates, proposes new search queries when nothing matches (the code runs them), explains verdicts under a validator that rejects any claim not present in the evidence, and, on request, judges whether the cited abstract supports the citing

sentence, quoting the abstract verbatim. The LLM never decides that a paper exists.

We make four contributions:

- **An evidence-first verifier** with a five-verdict taxonomy that separates fabricated from misattributed citations and from real citations with errors, and that answers UNVERIFIABLE rather than "fake" when coverage is thin. Every verdict carries an evidence bundle and a validated explanation, plus a corrected citation or real alternatives.
- **A grounded claim-support check** that catches a real paper cited for something it does not say, and that discards any judgement whose supporting quote is not verbatim in the abstract.
- **CiteBench and a held-out protocol.** Four splits of real, perturbed and fabricated citations sampled from two indexes; each split is scored once before its errors are studied.
- **An empirical comparison** against offline heuristics, a DOI-lookup baseline, and three LLM-only judges, with an ablation of the pipeline and an error analysis of every failure found.

On a fresh held-out split, CiteGuard detects every fake citation without a single false alarm on a real one, while LLM-only judges call 10–32% of correct citations fake (section 6). The system is released as a web application, a REST API and a command-line tool.

2 Related Work

Measuring citation hallucination. Studies of chatbot-written bibliographies report fabrication rates from about one in five to over half, depending on model and domain [2–4, 29]. Large-scale audits show the problem in published and preprint literature [30] and in citation URLs produced by deep-research agents [23]. These works measure the problem; we build a detector and measure its errors, especially false alarms on real citations.

Citation verifiers. Agrawal et al. [1] detect hallucinated references *without* external resources by asking the model consistency questions (for example, who the authors are) and observing that models answer inconsistently about references they invented. This is attractive when no index is available, but it inherits the model's coverage gaps. Retrieval-based verifiers appeared in 2026. HalluCiteChecker [24] is a lightweight offline toolkit; CiteCheck [12] retrieves candidates from arXiv, Crossref, Semantic Scholar and OpenAlex and lets an LLM score the citation–candidate pair into exact, minor and major labels; CiteTracer [16] cascades extraction, retrieval, deterministic field matching and class-specific LLM judges over a 12-code taxonomy; CiteAudit [25] uses a multi-agent pipeline and a large benchmark; and the AtomGPT checker [20] combines LLM field extraction with Semantic Scholar retrieval. HALLMARK [13] evaluates such verifiers under one protocol and argues that FPR limits deployment. CiteGuard shares the retrieval-first design of CiteCheck and CiteTracer but differs in three ways: the verdict is produced by calibrated linkage and rules rather than an LLM score, so the LLM cannot turn a non-match into a match; explanations are validated against the evidence before they are shown; and it checks claim support, which CiteCheck lists as out of scope. Our DOI-lookup baseline follows HALLMARK's.

Hallucination detection and grounded generation. General hallucination detectors compare samples for self-consistency [18] or decompose text into atomic facts checked against a knowledge source [19]. Retrieval-augmented generation grounds outputs in retrieved passages [15], and Gao et al. [6] evaluate whether generated citations support the generated statements. Our explanation validator applies the same idea narrowly: every reason must cite an evidence item, and every quoted value must occur in the evidence.

Claim verification. FEVER [26] and SciFact [28] frame verification as labelling a claim as supported or refuted by retrieved evidence. SemanticCite [8] applies this to citations with full-text retrieval and fine-tuned models. Our claim-support check is deliberately lighter: abstract-only, opt-in, and presented as a hint, with a verbatim-quote requirement that makes ungrounded judgements impossible to display.

Reference parsing and record linkage. Reference strings are parsed by CRF-based tools such as GROBID [17], whose accuracy depends on style and domain [27]. Matching a parsed reference to an index record is a record-linkage problem in the sense of Fellegi and Sunter [5]. We rely on open scholarly metadata from Crossref [9], OpenAlex [22] and Semantic Scholar [14].

3 Problem Formulation

The input is a document, a BibTeX file, a reference list or a single citation, optionally with the sentences that cite each reference. For each citation c, the system outputs a verdict $v(c) \in \{$VERIFIED, VERIFIED_WITH_ERRORS, MISATTRIBUTED, FABRICATED, UNVERIFIABLE$\}$, a confidence, a set of field-level discrepancy codes with the evidence behind each, an explanation, and a suggestion.

- VERIFIED: a real record matches and key fields agree.
- VERIFIED_WITH_ERRORS: a real record matches, but the year, venue, pages or identifier are wrong.
- MISATTRIBUTED: the title is real but the authors are substantially wrong, the DOI points to another work, or the title and authors come from different works (a chimera).
- FABRICATED: no plausible record exists in any healthy source although the citation has enough

metadata to find one, or its identifier is unregistered and nothing matches.
- UNVERIFIABLE: the evidence is insufficient, for example for grey literature, when sources are down, or when there is too little metadata.

We treat FABRICATED ∪ MISATTRIBUTED as the positive ("fake") class and report precision, recall and F1. The metric users feel most is the **false-positive rate on real citations**: the share of citations of real papers, with or without errors, that the system calls fake. Because a citation with a wrong year arguably "does not exist exactly as cited", we also report the FPR on *correct* citations only, which no reasonable verifier should flag. We further report five-class accuracy, macro-F1, the expected calibration error (ECE) [7] of the confidence, latency, and LLM tokens per citation.

4 The CiteGuard System

Figure 1 shows the pipeline. We describe each stage with its motivation and the design choice that keeps the LLM from deciding existence.

4.1 Ingestion, parsing and screening

Ingestion accepts PDF (through GROBID [17] when available, otherwise text extraction and reference-section segmentation), DOCX, LaTeX with `.bib`, BibTeX, plain text and single citations. In-text markers such as "[3]" or "(Smith, 2020)" are linked back to their references so that the citing sentence travels with the citation.

Parsing extracts DOIs, arXiv IDs, PMIDs and ISBNs by regular expression first; these identifiers are authoritative and no later step may change them. Fields are then parsed by GROBID, by a Unicode-aware heuristic parser for APA, IEEE, Vancouver and BibTeX, or, for references whose parse confidence is low, by the LLM. The LLM parser must transcribe faithfully and its title must occur (fuzzily) in the raw string; otherwise the next strategy is used.

Screening computes instant offline risk signals (an invalid DOI checksum or prefix, an arXiv ID whose date disagrees with the year, future dates, impossible pages, an unknown venue). These are shown immediately and feed the final confidence, but they never decide a verdict alone.

4.2 Retrieval and record linkage

Retrieval resolves identifiers at doi.org (which covers every registration agency) and arXiv, then searches Crossref, OpenAlex and Semantic Scholar, adding PubMed and arXiv by detected field. Each source sits behind a token-bucket rate limiter, retries with back-off, a circuit breaker, a cache (30 days for hits, 24 hours for misses) and a hard 20-second deadline per call. A slow source therefore lowers confidence instead of stalling the job.

Linkage normalises titles and names (NFKC, case and diacritic folding, LaTeX de-escaping; native-script and transliterated names; particles such as "van" and "dos") and computes field features: title similarity, first-author match, author-set precision and F1, year difference, venue similarity through an alias table, volume, issue and page agreement, and identifier agreement. A logistic model maps these features to $P(\text{same work})$. Candidates with $p \geq 0.9$ are accepted, those with $p < 0.5$ rejected, and the ambiguous band goes to the LLM adjudicator, which sees the citation and the top three candidates and may answer only with one of their IDs or "none"; a similarity floor rejects implausible picks. A chimera check searches by author and year to detect a real title paired with the authors of a different work.

Query reformulation runs only when nothing matches, and always before a FABRICATED verdict. The LLM proposes up to four alternative queries (without a subtitle, with transliterated names, author plus topic); the code runs them for at most two rounds, and the linkage model scores whatever they return. The LLM therefore only *surfaces candidates*.

4.3 Decision rules and parse recovery

Deterministic rules turn the evidence into a verdict. An unregistered DOI yields `ID_NOT_REGISTERED` and a search by metadata; a registered DOI whose title differs yields `ID_RESOLVES_TO_DIFFERENT_WORK`; a matched title whose listed authors are mostly absent from the record (author precision below 0.5) yields MISATTRIBUTED; and no match in at least two healthy relevant sources, for an article-like citation with enough metadata, yields FABRICATED, otherwise UNVERIFIABLE. Author checks use precision rather than F1, so a correct but truncated author list is a note (`AUTHORS_MISSING`), not a misattribution. The confidence of a FABRICATED verdict is $1 - P(\text{real work missed})$, using retrieval recall measured on the benchmark.

Parse recovery is a safety net for every parser. Before calling an unconfirmed citation fabricated, the verifier checks whether some retrieved record's title and first author occur verbatim in the raw string, as a whole segment between punctuation. If so, the parse was wrong: the reference is re-parsed around that title and verified again. The segment requirement stops a hallucination that *extends* a real title ("Attention Is All You Need for protein structure prediction") from being rescued.

4.4 Grounded LLM outputs

Explanations. Every verdict first receives a deterministic template explanation, so users see a reason immediately. In a second pass the LLM receives an evidence bundle (claimed fields, each source response with its status and URL, the best match, field differences, discrepancy codes) in which every item has

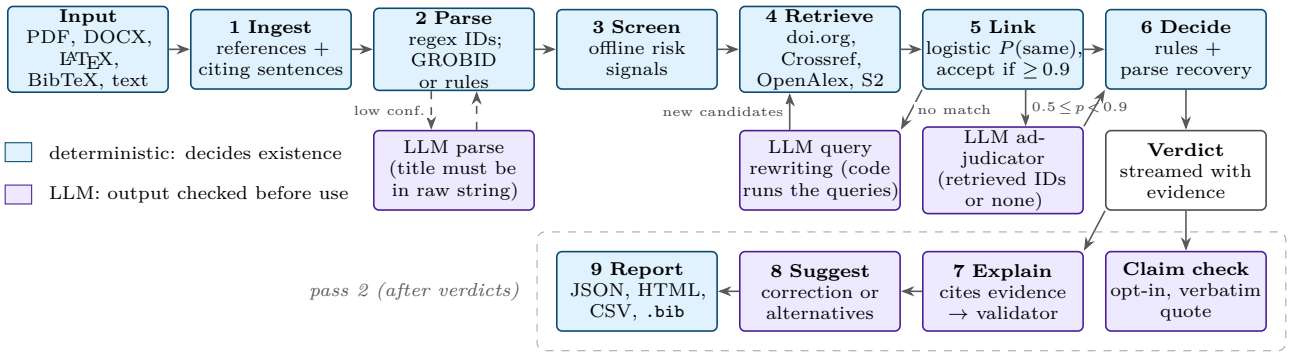


Figure 1: CiteGuard pipeline. Existence is decided only by blue, deterministic stages (retrieval, record linkage, rules). Purple stages use the LLM (Groq) for tasks whose output is checked: the parser's title must occur in the raw string, the adjudicator may only pick a retrieved candidate, reformulated queries are executed by code and re-scored, explanations must cite evidence IDs, and claim judgements must quote the abstract verbatim. Verdicts stream to the user after stage 6; explanations, suggestions and claim checks fill in during a second pass.

an ID, and returns strict JSON with a headline, reasons that cite evidence IDs, and a recommendation. A validator checks that each cited ID exists, that each quoted title, DOI, name or number occurs in the bundle, and that the headline does not call a citation fake unless the verdict is FABRICATED or MISATTRIBUTED. On failure the model gets one retry with feedback; after that the template explanation stays.

Suggestions. For real citations with errors, the matched record is formatted in the user's style with a field-level diff and a corrected BibTeX entry. For fabricated citations, candidates are retrieved by topic (and by the citing sentence when available) from Semantic Scholar and OpenAlex, and the LLM reranks them by abstract, returning up to three with a relevance note, or none if none fits.

Claim-support check (opt-in). For a verified citation with a citing sentence, CiteGuard fetches the abstract (from the retrieved records, then by DOI from OpenAlex or Semantic Scholar) and asks the LLM to label the sentence SUPPORTS, PARTIAL, NOT_SUPPORTED or CANNOT_DETERMINE, quoting the abstract. A judgement whose quote does not occur verbatim (after case and whitespace folding) is retried once and otherwise downgraded to CANNOT_DETERMINE. No LLM call is made when no abstract of at least 30 words exists. A NOT_SUPPORTED result adds the flag `CLAIM_NOT_SUPPORTED` and triggers a search for papers that do support the sentence, excluding the cited one.

4.5 Implementation

The backend is Python (FastAPI, asynchronous `httpx`, Pydantic), with PostgreSQL and Redis in production and SQLite in development; the frontend is React with TypeScript. A Docker Compose stack includes optional GROBID. All LLM calls go to Groq with strict JSON-schema outputs: `gpt-oss-20b` [21] parses and reformulates, and `gpt-oss-120b` adjudicates, explains, reranks and checks claims. A per-model limiter enforces the free-tier budget (8K tokens per minute, 200K per day); malformed outputs are retried once on the same model, then fall back to another model, and finally to deterministic behaviour. A check never fails because of the LLM, and with no API key the system still runs. The repository contains 110 backend and 7 frontend tests.

5 Experimental Setup

CiteBench. Each split has 150 items built by one recipe. *Real* (60): random journal articles from 1990–2024, sampled from both OpenAlex and Crossref so that ground truth does not come from a single index the system also queries, rendered in APA, IEEE, Vancouver or BibTeX, with the DOI kept in about half. *Perturbed* (60): year shifted by 2–3 years, venue swapped, or a non-existent DOI (labelled VERIFIED_WITH_ERRORS); authors swapped with another paper's, or another real paper's DOI (MISATTRIBUTED); and an invented title carrying a real DOI, a typical LLM hallucination (FABRICATED). *Fabricated* (30): invented but plausible titles with real-looking authors and real venues (FABRICATED). Labels follow from construction.

Held-out protocol. We use four splits with disjoint DOIs: `dev` (seed 7) for development, and `test` (seed 23), `test2` (seed 41) and `test3` (seed 59), each scored once before anyone looked at its errors. After its first score, a split's errors were analysed and fixed, so later scores on it are optimistic and we do not report them as estimates. `test3` was built for this paper and scored once with the final code; it is our main result.

Systems compared.

- **Offline heuristics**: the screening signals alone, thresholded at 0.3.
- **DOI lookup** (HALLMARK's lower-bound baseline [13]): a citation is flagged only if it has a DOI that doi.org does not resolve; citations without a DOI are not flagged.

- **LLM-only**: the model is asked whether the publication "really exists exactly as cited, using only your own knowledge" and answers yes, no or unsure (mapped to VERIFIED, FABRICATED, UNVERIFIABLE), with no retrieval and no fallback to another model. We test `gpt-oss-20b`, `qwen3.8-27b` and `gpt-oss-120b`, all served by Groq.
- **CiteGuard**: retrieval with exact identifier and title matching only; the full deterministic pipeline (`no_llm`); and the full pipeline with the Groq LLM (`full`).

6 Results

6.1 Main comparison on the fresh held-out split

Table 1 answers the central question: does grounding the decision in retrieval beat asking an LLM? Three findings stand out.

Retrieval-grounded verification reaches a far better operating point. Every CiteGuard variant detects all 60 fake citations (recall 1.000, precision 1.000) without calling a single real citation fake. No baseline comes close: the best LLM-only judge, `gpt-oss-120b`, reaches a fake F1 of 0.727 but calls 13.3% of the perfectly correct citations fake (and 26.7% of all real ones); `qwen3.8-27b` calls 31.7% of correct citations fake.

A larger model is not a safer judge. `gpt-oss-20b` has a lower false-positive rate on correct citations (10.0%) than `gpt-oss-120b` (13.3%); the larger model mostly answers "unsure" (40 of 60 correct citations), which is honest but not useful. `qwen3.8-27b` also failed to return valid structured output for 27 of 150 citations even after a retry; these count as "unsure".

Identifier checks alone miss fabrication entirely. The DOI-lookup baseline detects none of the 60 fakes: CiteBench's invented titles either carry no DOI or borrow a real one, which resolves. Its only flags are the 10 real papers cited with a non-existent DOI, which are real papers with an error, so they count as false positives. The offline heuristics flag 2 of 60 fakes.

The LLM changes no verdict on this split (its predictions are identical to `no_llm`) at 241 tokens per citation. All remaining CiteGuard errors are six correct citations labelled VERIFIED_WITH_ERRORS, analysed in section 6.6.

6.2 What the LLM-only judges get wrong

Table 2 breaks the LLM-only answers down by gold group. The judges catch most fabrications (30–37 of 40), but mainly because they distrust what they do not recognise; the same distrust makes them call 6–19 of 60 correct citations fake. For real papers cited with a small error (a year off by two, a swapped venue), they mostly answer "fake" (16–22 of 30). Under a prompt that asks whether a work exists "exactly as cited" this answer is defensible, which is why we report the FPR on correct citations separately; but it shows that a yes/no judge cannot tell a typo from a fabrication. A user receives no corrected citation, no indication of which field is wrong, and no evidence to check, which are exactly the outputs CiteGuard's five verdicts and discrepancy codes provide.

6.3 Earlier held-out estimates

Table 3 lists the first score of every held-out split. The code improved between them: `test` exposed Vancouver author lists the parser could not read, `test2` a handful of smaller parsing gaps (section 6.6), and `test3` measures the final code. Fake recall was 1.000 on every split. The remaining errors were all on real citations: called fake on `test` (5 of 90) and `test2` (1 of 90), and only given a minor-error verdict on `test3`. The false-positive rate fell from 5.6% to 1.1% to 0% as parsing failures were fixed.

6.4 Ablation

Table 4 adds components one at a time on `dev`. Offline heuristics alone find no fakes: a fabricated citation with a real venue and a plausible title has no surface defect. Retrieval is the component that matters. On these tidy synthetic citations exact matching already solves `dev`, and fuzzy linkage, the full rules and the LLM change nothing measurable. The held-out `test3` split confirms this rather than contradicting it: exact matching scores 0.967 accuracy against 0.960 for the full deterministic pipeline (a one-citation difference), and both detect every fake with no false alarm (table 1). The improvements that mattered on the held-out splits were in *parsing* and in the decision rules, which both rungs share (section 6.6). **We therefore found no benchmark evidence that fuzzy linkage or the LLM improves verdicts on synthetic citations.** Both are designed for inputs CiteBench does not contain (typos in titles, transliterated names, references a chatbot half-remembers); measuring them needs the chatbot-written split described in section 7. The LLM's measurable contribution here is in explanations and claim checks, evaluated next.

6.5 Explanations and claim support

Grounded explanations. On 30 `dev` citations run with explanations, all 19 non-verified citations received an LLM explanation, and all 19 passed the grounding validator (100%; one retry allowed), at 1,972 tokens per citation. Verified citations keep the template explanation by design. The validator caught one real defect during development: for a correctly UNVERIFIABLE citation, the LLM wrote "Could not verify: citation appears fabricated", which led to the

Table 1: Main result: fresh held-out split `test3` (150 citations, scored once). Fake = FABRICATED ∪ MISATTRIBUTED. "FPR correct" is the share of the 60 correct citations called fake; "FPR real" adds the 30 real citations with errors. LLM-only baselines cannot output VERIFIED_WITH_ERRORS or MISATTRIBUTED, which caps their five-class accuracy; compare them on the binary columns.

	Five-class		Fake detection			FPR (%) ↓		LLM
System	Acc. ↑	M-F1 ↑	P ↑	R ↑	F1 ↑	correct	real	tok./cit.
Offline heuristics only	0.400	0.145	**1.000**	0.050	0.095	**0.0**	**0.0**	0
DOI lookup only [13]	0.400	0.150	–	0.000	0.000	**0.0**	11.1	0
LLM-only, `gpt-oss-20b`	0.473	0.323	0.632	0.800	0.706	10.0	31.1	784
LLM-only, `qwen3.8-27b`	0.367	0.264	0.506	0.700	0.587	31.7	45.6	220
LLM-only, `gpt-oss-120b`	0.327	0.244	0.667	0.800	0.727	13.3	26.7	671
CiteGuard, exact matching only	**0.967**	**0.970**	**1.000**	**1.000**	**1.000**	**0.0**	**0.0**	0
CiteGuard, deterministic (`no_llm`)	0.960	0.964	**1.000**	**1.000**	**1.000**	**0.0**	**0.0**	0
CiteGuard, full (Groq LLM)	0.960	0.964	**1.000**	**1.000**	**1.000**	**0.0**	**0.0**	241

Table 2: How the LLM-only judges answered on `test3`: counts of "exists" / "fake" / "unsure" per gold group. Every "fake" in the first two rows is a false alarm on a real paper.

	`gpt-oss-20b`			`qwen3.8-27b`			`gpt-oss-120b`		
Gold group	ex.	fake	uns.	ex.	fake	uns.	ex.	fake	uns.
Correct (60)	34	6	20	25	19	16	12	8	40
Real w/ errors (30)	4	22	4	2	22	6	2	16	12
Misattributed (20)	5	11	4	1	12	7	2	11	7
Fabricated (40)	1	37	2	0	30	10	0	37	3

Table 3: First (held-out) scores of the deterministic pipeline on each split, each with the code of its time. Latency is p50 / p95 seconds per citation; `test3` ran on a source cache already filled by the exact-matching run, so its latency is not comparable.

Split	Acc.	M-F1	P	R	FPR%	ECE	Lat. (s)
`test` (seed 23)	0.960	0.960	0.923	1.000	5.6	0.058	1.3 / 24
`test2` (seed 41)	0.987	0.986	0.984	1.000	1.1	0.061	1.2 / 20
`test3` (seed 59)	0.960	0.964	1.000	1.000	0.0	0.013	cached

rule that a headline may call a citation fake only if the verdict says so.

Claim support. For 25 real `test2` papers with abstracts, `gpt-oss-120b` judged two claims each: the paper's own concluding sentence with lead-ins such as "We show that" removed (should be supported), and the concluding sentence of a different, random paper (should not be). All 25 own claims were labelled SUPPORTS (0 false alarms), and 23 of 25 foreign claims were labelled NOT_SUPPORTED (92%); the other two were vague sentences that received the cautious CANNOT_DETERMINE. The grounding check dropped one judgement whose quote was not in the abstract. A check costs about 1.6K tokens uncached. Synthetic claims measure false alarms and gross topic mismatches, not subtle misuse such as overgeneralising a finding; that needs a hand-labelled set.

Self-check on this paper's bibliography. We ran CiteGuard on the 30 entries of this paper's own reference list (appendix A). It verified 28 and flagged two as VERIFIED_WITH_ERRORS; none was called fake. Both flags are false alarms caused by how indexes record a work: Crossref lists a 2009 proceedings paper under its series (*Lecture Notes in Computer Science*) rather than its proceedings title, and Semantic Scholar gives an arXiv record's "volume" as `abs/1706.04599`. Such metadata conventions are the main source of minor false alarms we observed, and they are harmless by design: a VERIFIED_WITH_ERRORS verdict shows the user both values.

6.6 Error analysis

Every error found on `dev`, `test` and `test2` was traced to a cause and fixed with a regression test (table 5). The six `test3` errors were analysed but left unfixed, so that `test3` remains a clean held-out estimate. All six are correct citations labelled VERIFIED_WITH_ERRORS, a minor error that still shows the user the matched record: a title containing "II." that the sentence splitter cut; an IEEE title containing "(1998)" whose parenthesised year was taken for the publication year (2005); a Japanese record with ALL-CAPS names and the volume "2017.23"; and three citations rendered from garbled index metadata (for example the author "PG Student, D. o. S. E. C. c. J. H. I."), where the record itself is the problem.

Two patterns stand out. First, almost every false alarm began as a *parsing* failure: the remainder of an unreadable author list became the "title", nothing matched, and a real paper was called fabricated with high confidence. Parse recovery (section 4) addresses this class generally rather than one name at a time. Second, fixes interact: changing author fabrication from F1 to precision broke two `dev` items, which the regression runs caught before the change shipped. Re-running all earlier splits after each fix was essential.

Defects found by the demo. Recording the demo exposed two LLM-side issues that no benchmark met-

Table 4: Ablation ladder on `dev` (150 citations; development split, not held out). Each row adds one component. `full` and LLM-only were measured before the parse-recovery fix.

	Five-class		Fake detection			FPR (%)↓		LLM
System	Acc.↑	M-F1↑	P↑	R↑	F1↑	correct	real	tok./cit.
Offline heuristics only	0.400	0.143	–	0.000	0.000	0.0	0.0	0
+ retrieval, exact matching	1.000	1.000	1.000	1.000	1.000	0.0	0.0	0
+ fuzzy linkage and full rules (`no_llm`)	1.000	1.000	1.000	1.000	1.000	0.0	0.0	0
+ Groq LLM (`full`)	1.000	1.000	1.000	1.000	1.000	0.0	0.0	cached
LLM-only, `gpt-oss-120b`	0.340	0.249	0.595	0.783	0.676	15.0	35.6	661

Table 5: Causes of the errors found by the held-out splits and the Docker smoke test, and their fixes. "Real→fake" are false alarms.

Found on	Cause	Effect
`dev`	ALL-CAPS Vancouver titles swallowed the journal	real→unverif.
`dev`	Non-Latin names folded to empty strings; CJK family-name order	real→fake
`dev`	Quoted phrase inside a title taken as an IEEE title	real→unverif.
`test`	Vancouver authors such as "Van Tinteren H", "Helland Å", Unicode hyphens	real→fake (5)
`test`	APA volume ranges read as pages	minor error
`test2`	"St. Petersburg" split as a sentence end	real→fake
`test2`	Brazilian initials ("BVdS"); partial family names in Crossref	parse failure
smoke	Correct but truncated author list judged by F1	real→misattr.

ric showed. The explainer once returned the headline "Real paper with errors:" with nothing after the label; the validator now rejects a label-only headline and asks again (with a regression test). In the same explanation, one reason inferred that the citation "may have been assembled from mixed sources". It cited real evidence IDs and quoted nothing false, so it passed the validator, but the inference was not warranted. The validator checks values and evidence IDs, not reasoning, which is a limit of string-level grounding.

6.7 Cost and latency

The deterministic pipeline uses no LLM tokens. Its median latency is about one second per uncached citation; the 95th percentile (about 20 seconds) is set by occasional source time-outs, bounded by the per-call deadline. On the Groq free tier, verdicts in `full` mode cost 241 tokens per citation on `test3`, and explanations about 2K; explanation latency (19 s median) is dominated by the tokens-per-minute limit rather than by the pipeline, and verdicts stream before explanations.

7 Discussion and Limitations

Why evidence first works. The LLM-only judges fail in a predictable direction: they confuse "I do not recognise this paper" with "this paper does not exist". Retrieval reverses the burden of proof: a real paper is found regardless of how obscure it is, and a fabricated one is not found regardless of how plausible it sounds. Separating FABRICATED from UNVERIFIABLE keeps thin coverage from turning into false accusations, and keeping the LLM to checkable tasks means its errors are caught by validators instead of reaching the verdict.

Limitations.

- **Synthetic benchmark.** CiteBench citations are tidy and the fabricated titles are template-generated, not chatbot-written; the real split contains only journal articles. Perfect or near-perfect scores mean the benchmark is too easy for retrieval-based systems, not that the problem is solved. A chatbot-written split with manual labels and hard negatives (books, theses, non-English and very recent work) is the most important next step.
- **Comparability.** We compare against baseline *types* from the literature (DOI lookup, zero-shot LLM judges) on our benchmark, not against the released systems of Khajavi et al. [12], Li et al. [16] or Kim et al. [13] on theirs; the numbers are not directly comparable across papers.
- **Small samples.** Splits of 150 give wide confidence intervals (one error is 0.7 points of accuracy), and the explanation and claim studies have 19 and 50 judgements.
- **Coverage.** Verification is only as good as the indexes. Field-specific databases, books and grey literature are thinly covered, which CiteGuard reports as UNVERIFIABLE.
- **Claim support is abstract-only**, so it cannot see support that appears only in the full text, and it is shown as a hint, not a verdict.
- **String-level grounding.** The explanation validator verifies evidence IDs and quoted values, not the soundness of an inference drawn from them.
- **LLM-only prompt.** Prompt wording shifts LLM-only FPR substantially [13]; we used one prompt per baseline.

8 Conclusion

Fabricated citations are a hallucination problem that should not be solved with more hallucination. CiteGuard decides whether a cited work exists by retrieval

and deterministic record linkage, and uses an LLM only where its output can be checked against evidence: parsing, adjudicating among retrieved candidates, reformulating searches, explaining verdicts, and judging claim support with verbatim quotes. On a fresh held-out split it detects every fake citation and calls no real citation fake, whereas LLM-only judges call 10–32% of correct citations fake and a DOI check misses fabrications entirely. Future work is a chatbot-written benchmark with human labels, hard negatives, and a hand-labelled claim-misuse set.

Use of AI assistants. AI assistants were used for parts of the coding and drafting. All experiments were run with the scripts in the repository, every number in the result tables is generated from the run outputs (`paper/make_results.py`), and every reference was verified (appendix A).

References

[1] Ayush Agrawal, Mirac Suzgun, Lester Mackey, and Adam Tauman Kalai. Do language models know when they're hallucinating references? In *Findings of the Association for Computational Linguistics: EACL 2024*, pages 912–928, 2024. doi: 10.18653/v1/2024.findings-eacl.62.

[2] Hussam Alkaissi and Samy I. McFarlane. Artificial hallucinations in ChatGPT: Implications in scientific writing. *Cureus*, 2023. doi: 10.7759/cureus.35179.

[3] Mehul Bhattacharyya, Valerie M. Miller, Debjani Bhattacharyya, and Larry E. Miller. High rates of fabricated and inaccurate references in ChatGPT-generated medical content. *Cureus*, 2023. doi: 10.7759/cureus.39238.

[4] Mikaël Chelli, Jules Descamps, Vincent Lavoué, Christophe Trojani, Michel Azar, Marcel Deckert, Jean-Luc Raynier, Gilles Clowez, Pascal Boileau, and Caroline Ruetsch-Chelli. Hallucination rates and reference accuracy of ChatGPT and Bard for systematic reviews: Comparative analysis. *Journal of Medical Internet Research*, 26:e53164, 2024. doi: 10.2196/53164.

[5] Ivan P. Fellegi and Alan B. Sunter. A theory for record linkage. *Journal of the American Statistical Association*, 64(328):1183–1210, 1969. doi: 10.1080/01621459.1969.10501049.

[6] Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. Enabling large language models to generate text with citations. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 6465–6488, 2023. doi: 10.18653/v1/2023.emnlp-main.398.

[7] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330, 2017.

[8] Sebastian Haan. SemanticCite: Citation verification with AI-powered full-text analysis and evidence-based reasoning, 2025. arXiv:2511.16198.

[9] Ginny Hendricks, Dominika Tkaczyk, Jennifer Lin, and Patricia Feeney. Crossref: The sustainable source of community-owned scholarly metadata. *Quantitative Science Studies*, 1(1):414–427, 2020. doi: 10.1162/qss_a_00022.

[10] Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, and Ting Liu. A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *ACM Transactions on Information Systems*, 43(2):1–55, 2025. doi: 10.1145/3703155.

[11] Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):1–38, 2023. doi: 10.1145/3571730.

[12] Khashayar Khajavi, Shaghayegh Sadeghi, Rise Adhikari, and Alexander Tessier. CiteCheck: Retrieval-grounded detection of LLM citation hallucinations in scientific text, 2026. arXiv:2605.27700.

[13] Kayoon Kim, Wieland Brendel, and Patrik Reizinger. HALLMARK: How to diagnose LLM citation verifiers and when to deploy them, 2026. arXiv:2607.18360.

[14] Rodney Kinney et al. The Semantic Scholar open data platform, 2023. arXiv:2301.10140.

[15] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474, 2020.

[16] Mingzhe Li, Zhiqiang Lin, and Shiqing Ma. Source or it didn't happen: A multi-agent framework for citation hallucination detection, 2026. arXiv:2605.08583.

[17] Patrice Lopez. GROBID: Combining automatic bibliographic data recognition and term extraction for scholarship publications. In *Research and Advanced Technology for Digital Libraries (ECDL 2009)*, volume 5714 of *Lecture Notes in Computer Science*, pages 473–474, 2009. doi: 10.1007/978-3-642-04346-8_62.

[18] Potsawee Manakul, Adian Liusie, and Mark Gales. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 9004–9017, 2023. doi: 10.18653/v1/2023.emnlp-main.557.

[19] Sewon Min, Kalpesh Krishna, Xinxi Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. FActScore: Fine-grained atomic evaluation of factual precision

in long form text generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12076–12100, 2023. doi: 10.18653/v1/2023.emnlp-main.741.

[20] Harichandana Neralla, Jaehyung Lee, Aldo H. Romero, and Kamal Choudhary. Hallucination detector: A hybrid LLM and Semantic Scholar tool calling for detecting hallucination in scientific literature on AtomGPT.org, 2026. arXiv:2607.09774.

[21] OpenAI. gpt-oss-120b & gpt-oss-20b model card, 2025. arXiv:2508.10925.

[22] Jason Priem, Heather Piwowar, and Richard Orr. OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts, 2022. arXiv:2205.01833.

[23] Delip Rao, Eric Wong, and Chris Callison-Burch. Detecting and correcting reference hallucinations in commercial LLMs and deep research agents, 2026. arXiv:2604.03173.

[24] Yusuke Sakai, Hidetaka Kamigaito, and Taro Watanabe. HalluCiteChecker: A lightweight toolkit for hallucinated citation detection and verification in the era of AI scientists, 2026. arXiv:2604.26835.

[25] Kaiwen Shi, Weixiang Sun, Zheyuan Zhang, Lichao Sun, Nitesh V. Chawla, and Yanfang Ye. CiteAudit: You cited it, but did you read it? A benchmark for verifying scientific references in the LLM era, 2026. arXiv:2602.23452.

[26] James Thorne, Andreas Vlachos, Christos Christodoulopoulos, and Arpit Mittal. FEVER: a large-scale dataset for fact extraction and VERification. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, pages 809–819, 2018. doi: 10.18653/v1/N18-1074.

[27] Dominika Tkaczyk, Andrew Collins, Paraic Sheridan, and Joeran Beel. Machine learning vs. rules and out-of-the-box vs. retrained: An evaluation of open-source bibliographic reference and citation parsers. In *Proceedings of the 18th ACM/IEEE on Joint Conference on Digital Libraries*, pages 99–108, 2018. doi: 10.1145/3197026.3197048.

[28] David Wadden, Shanchuan Lin, Kyle Lo, Lucy Lu Wang, Madeleine van Zuylen, Arman Cohan, and Hannaneh Hajishirzi. Fact or fiction: Verifying scientific claims. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 7534–7550, 2020. doi: 10.18653/v1/2020.emnlp-main.609.

[29] William H. Walters and Esther Isabelle Wilder. Fabrication and errors in the bibliographic citations generated by ChatGPT. *Scientific Reports*, 13(1):14045, 2023. doi: 10.1038/s41598-023-41032-5.

[30] Zhenyue Zhao, Yihe Wang, Toby Stuart, Mathijs De Vaan, Paul Ginsparg, and Yian Yin. LLM hallucinations in the wild: Large-scale evidence from non-existent citations, 2026. arXiv:2605.07723.

A CiteGuard run on this paper's bibliography

Key	Verdict	Confidence	Matched in
walters2023fabrication	verified	0.99	crossref
bhattacharyya2023high	verified	0.99	crossref
alkaissi2023artificial	verified	0.99	crossref
chelli2024hallucination	verified	0.99	crossref
zhao2026wild	verified	0.95	arxiv
rao2026detecting	verified	0.95	arxiv
agrawal2024hallucinating	verified	0.99	crossref
kim2026hallmark	verified	0.95	arxiv
khajavi2026citecheck	verified	0.95	arxiv
li2026source	verified	0.99	arxiv
shi2026citeaudit	verified	0.99	arxiv
sakai2026hallucitechecker	verified	0.95	arxiv
neralla2026hallucination	verified	0.95	arxiv
haan2025semanticcite	verified	0.95	arxiv
ji2023survey	verified	0.99	crossref
huang2025survey	verified	0.99	crossref
lewis2020retrieval	verified	0.99	semantic_scholar
manakul2023selfcheckgpt	verified	0.99	crossref
min2023factscore	verified	0.99	crossref
gao2023enabling	verified	0.99	crossref
thorne2018fever	verified	0.99	crossref
wadden2020fact	verified	0.99	crossref
guo2017calibration	verified with errors	0.99	semantic_scholar
openai2025gptoss	verified	0.95	arxiv
lopez2009grobid	verified with errors	0.99	crossref
tkaczyk2018machine	verified	0.99	openalex
fellegi1969theory	verified	0.99	crossref
hendricks2020crossref	verified	0.99	crossref
priem2022openalex	verified	0.95	arxiv
kinney2023semantic	verified	0.95	arxiv

Generated by `paper/verify_references.py` (deterministic mode, no LLM). The two *verified with errors* verdicts are discussed in section 6.

B Reproducing the results

```
cd backend
D=../eval/datasets/citebench_test3.jsonl
python -m citeguard.eval.build --real 60 --perturbed 60 --fabricated 30 \
    --seed 59 --out $D
python -m citeguard.eval.run --dataset $D --mode no_llm      # also: heuristics,
python -m citeguard.eval.run --dataset $D --mode full        #   doi_only, exact
python -m citeguard.eval.run --dataset $D --mode llm_only \
    --llm-model openai/gpt-oss-120b --name citebench_test3-llm_only-gpt-oss-120b
python -m citeguard.eval.claims --dataset ../eval/datasets/citebench_test2.jsonl --n 25
python ../paper/verify_references.py
python3 ../paper/make_results.py && cd ../paper && tectonic main.tex
```